

altairsim — Writing Altair cassettes — the MITS Tapes disk

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A CP/M disk full of the **MITS distribution tapes** — every 4K, 8K, Extended and Disk BASIC, and the Programming System II monitor — as `.TAP` images, together with **WRTAPE.COM**, the utility that writes one of them back out through the cassette port the way a real Altair did.

Boot it, put a blank tape in the recorder, and record a bootable cassette you can then load on the `basic4k` / `basic8k` / `ps2` machines — or on real hardware, from the `.WAV` it makes.

The disk and WRTAPE are **Mike Douglas's**, from deramp.com. This directory adds a machine file that plugs an 88-ACR into the bus alongside the disk, and the host-bridge utilities (`HDIR`, `R`, `W`) so you can move `.TAP` files in and out of your working directory.

From this directory (`acr`):

```
altairsim mitstapes.toml

48K CP/M
Version 2.2mits (07/28/80)
A>
```

`TYPE CONTENTS.TXT` at the `A>` prompt is the disk's own annotated tape list; it is also reproduced below.

Recording a tape

WRTAPE reads a `.TAP` off the disk and writes it, byte by byte, to the 88-ACR. On a real Altair the operator held `PLAY+RECORD` on the deck first; here that is one `MOUNT` with `mode=record`, and `CREATE` makes the output file since it does not exist yet. `MOUNT` is a **monitor** command, so you reach it with `^E`:

The `A>` prompt is CP/M's; the `altairsim>` prompt is the monitor's. `^E` is a keypress, not a typed command — it toggles from CP/M to the monitor.

```
$ altairsim mitstapes.toml           # boots to A>

A>^E                                # press ^E -> the monitor
altairsim> MOUNT acr0:tape "8kbas40.wav" CREATE mode=record # blank tape in, PLAY+RECORD down
altairsim> RUN                        # resume CP/M -> back at A>

A>WRTAPE 8KBAS40.TAP                # press RETURN; wait for:
                                     #   File Transfer Complete

A>^E                                # press ^E -> the monitor
altairsim> UNMOUNT acr0:tape         # flush the recording to disk
```

That leaves `8kbas40.wav` in this directory — a **300-baud FSK audio cassette**, 44.1 kHz, that decodes back with zero framing errors and plays into a real Altair's 88-ACR. Name the file `.tap` instead of `.wav` and you get the raw **byte image** rather than audio; everything else is the same.

- **A fresh tape does not need a REWIND.** `CREATE` starts an empty tape with the head at the beginning. (You only REWIND when you are recording over a tape that already has something on it.)
- **The transfer is instant at the default clock.** On real hardware it runs at 30 cps — about two minutes for 4K BASIC, four and a half for 8K. `SET acr0:tape rate=real` before you record buys back that true speed, and the console tape counter climbs while it runs.
- **A second cassette port.** `WRTAPE` writes through the 88-SIO at 6/7 (the Altair Cassette port, which is where the `acr0` in this machine lives) by default. Following the file name with a `2` — `WRTAPE 8KBAS40.TAP 2` — writes through the second port of an 88-2SIO instead, for a machine strapped that way.

Cassette and paper-tape images are identical, so the exact same steps write a paper tape.

Moving tapes in and out — the host bridge

The `default` machine this builds on carries an 88-hostbridge (`hb0`), and the disk has its three guest utilities installed:

These are ordinary CP/M programs, typed at the `A>` prompt:

```
A>HDIR          # list your working directory (the host)
A>R MYPROG.TAP  # host -> CP/M: copy a .TAP onto drive A:
A>W 8KBAS40.TAP # CP/M -> host: copy a .TAP back out to your working directory
```

So the full loop is: **R** a tape image you got from somewhere onto the disk, then **WRTAPE** it to a cassette — or **W** one of the disk's own tapes out to keep it.

What is on the disk

Boot tapes — each loads and runs on its own through the bootstrap MITS shipped:

File	What it is
4KBAS32.TAP	4K BASIC ver 3.2
4KBAS40.TAP	4K BASIC ver 4.0
8KBAS32.TAP	8K BASIC ver 3.2
8KBAS40.TAP	8K BASIC ver 4.0
EXTBAS40.TAP	Extended BASIC ver 4.0
EXTBAS41.TAP	Extended BASIC ver 4.1
DSKBAS50.TAP	Disk BASIC 5.0
PS2-MON.TAP	MITS Programming System II monitor
DBL40.TAP	Disk Boot Loader ver 4.0
DBL45.TAP	Disk Boot Loader ver 4.5

Boot tapes strapped for port B of an 88-2SIO as the cassette input:

File	What it is
8K2SIO.TAP	8K BASIC ver 4.0
EXT2SIO.TAP	Extended BASIC ver 4.1
PS2-2SIO.TAP	MITS Programming System II monitor

Programming System II files:

File	What it is
PS2-EDT.BIN	Editor
PS2-ASM.BIN	Assembler (overlays the editor)
PS2-AM2.BIN	Assembler (co-resident with the editor)
PS2-DBG.BIN	Debugger

The disk also holds `WRTAPE.COM` and its source `WRTAPE.ASM`, `CONTENTS.TXT`, the host-bridge utilities `R.COM` / `W.COM` / `HDIR.COM`, and the usual CP/M tools (`PIP`, `STAT`, `DDT`, `ASM`, `SUBMIT`, `FORMAT`, and so on).

The files in this directory

File	What it is
<code>mitstapes.toml</code>	The machine: <code>base = "default"</code> plus the MITS Tapes disk in drive A: and an 88-ACR at port 006.
<code>MITS Tapes.dsk</code>	Mike Douglas's disk, with the host-bridge utilities added.
<code>README.md</code>	This file.
<code>README.pdf</code>	This file, built by CI.

The recorder starts **empty** — a deck with no cassette in it — so a plain boot writes nothing; you `MOUNT` a tape when you are ready to record, as above.